

Huawei CloudEngine S5735-S Series Switches

Performance Evaluation and Feature Validation

Executive Summary

Huawei CloudEngine S5735-S series switches are standard gigabit switches with GbE access ports and 10GbE uplink ports. They are ideal to be deployed in the access or aggregation layer of campus networks, and access layer of data center networks.

Tolly engineers evaluated Huawei CloudEngine S5735-S series switches' performance and validated their features. Built upon Huawei's high performance Versatile Routing Platform (VRP) software, CloudEngine S5735-S series switches support flexible Ethernet networking, various security features, and enhanced IPv4 / IPv6 Layer 3 features.

The Bottom Line

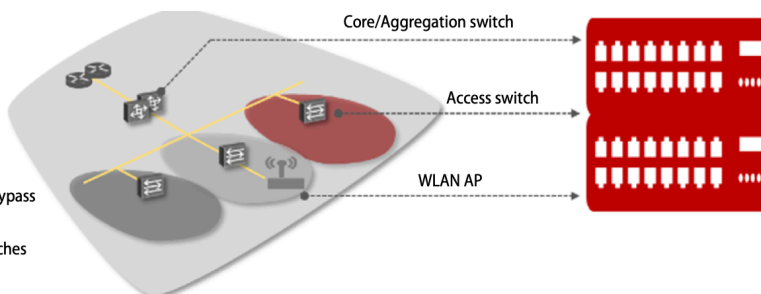
Huawei CloudEngine S5735-S Series Switches:

- 1 Support Huawei's Super Virtual Fabric (SVF) technology, which virtualizes core/aggregation devices (parent) and access devices (clients) into one logical device for easier management. SVF clients can include two layers of access switches (ASes) and one layer of WLAN APs, with ASes supporting stacked devices
- 2 Support common Layer 2 protocols and enhanced Layer 3 features
- 3 Support numerous security features including dynamic ARP inspection, IP source guard, PPPoE+, secure boot, and more

Huawei Super Virtual Fabric (SVF)

Typical Campus Network

- Layered tree architecture
- Horizontal stacking of core/aggregation devices
- WLAN ACs connect to core/aggregation devices in bypass mode
- High quantities of access switches



Converged Campus Network

- Cloud architecture, on-demand expansion
- Campus network virtualized into **one** device
- Access switch = WLAN AP
- Aggregation switch = WLAN AC

1. **Simplified Management:** The entire network is virtualized into one device. Huawei eSight, a unified network management platform, only needs to manage one network element, supporting plug-and-play.
2. **Centralized Control:** ACL, QoS, and user security policies are centrally controlled by the core device.
3. **Reduced Investment:** Low-specification fixed switches can be used as access devices, reducing the overall investment on devices.

Source: Tolly, November 2019

Figure 1

Huawei CloudEngine S5735-S-V2 Series Switches

Performance Evaluation and Feature Validation

Executive Summary

The CloudEngine S5735-S-V2 series switch is based on a new generation of high-performance hardware and Huawei's unified software platform. It features enhanced Layer 3 capabilities, easy operation and maintenance, flexible Ethernet networking, mature IPv6 features, and is widely used in various application scenarios such as enterprise campus access and aggregation, data center access, and more.

Tolly engineers validated the specifications and functions of the Huawei CloudEngine S5735-S-V2 series switch.

The Bottom Line

Huawei CloudEngine S5735-S-V2 series switches:

- 1 Support GE/2.5GE/Multi-GE electrical downlink ports and 10GE optical uplink ports. The access rate can be flexibly selected to meet the access requirements of endpoints with various rates, enabling flexible networking
- 2 Support various security features such as CPU defense, IP Source Guard (IPSG), Dynamic ARP Inspection (DAI), Source Address Validation Improvements (SAVI), MACsec and trusted boot

Small- and Medium Business Network

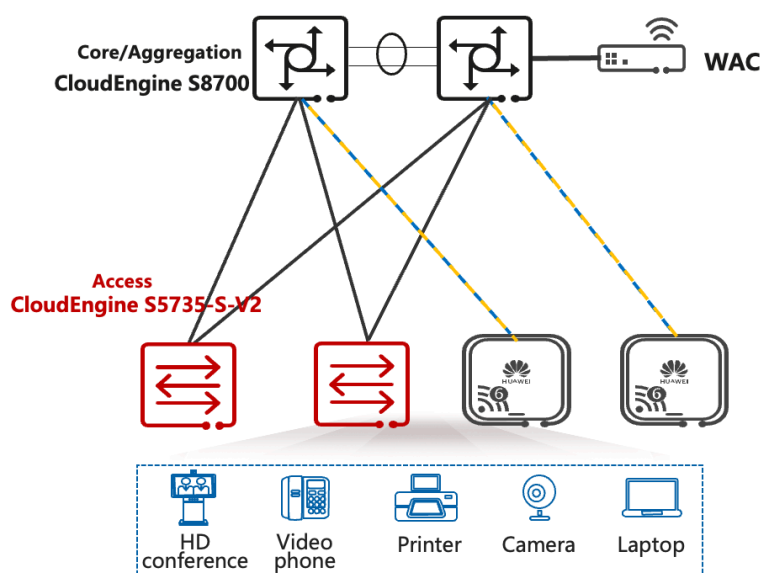


Figure 1

Report Link: <https://www.tolly.com/publications/detail/224127>



Test Results

Tolly engineers conducted tests on the Huawei CloudEngine S5735-S-V2 series switches (hereinafter referred to as the S5735-S-V2 switch) to assess their functionality and performance.

Test results apply to all CloudEngine S5735-S-V2 models including:

S5735-S24T4XE-V2,
S5735-S24P4XE-V2,
S5735-S24U4XE-V2,
S5735-S48T4XE-V2,
S5735-S48P4XE-V2,
S5735-S48U4XE-V2,
S5735-S24ST4XE-V2,
S5735-S48S4XE-V2,
S5735-S24HS4XE-V2,
S5735-S48HS4XE-V2,
S5735-S24PN4XE-V2,
S5735-S48PN4XE-V2,
S5735-S24T4XE-C-V2,
S5735-S48T4XE-C-V2.

A list of verified features is provided in Tables 1 and 2 on Pages 6 and 7. Detailed results are as follows.

Ports

Different models of the S5735-S-V2 support varying numbers and types of ports. Tolly engineers verified that the GE ports, Multi-GE ports (up to 2.5GE), and 10GE ports support line-rate forwarding when interconnected with ports of the same type.

Capacity

MAC Table

The S5735-S-V2 switch's MAC table supports 32K (32,768) addresses. Traffic matching any MAC address in the MAC table was forwarded without packet loss and without broadcasting.

ARP Table

The S5735-S-V2 switch's ARP table supports 4,000 ARP entries. Traffic matching these entries was forwarded without packet loss.

ND Table

The S5735-S-V2 switch's ND table supports 3,072 ND entries. Traffic matching these entries was forwarded without packet loss.

Routing Table/FIB

The S5735-S-V2 switch's IPv4 routing table and FIB (Forwarding Information Base) table each supports 8,192 IPv4 routes. Traffic matching all routes in the FIBv4 table was forwarded without packet loss.

The S5735-S-V2 switch's IPv6 routing table and FIBv6 table each supports 3,072 IPv6 routes. Traffic matching all routes in the FIBv6 table was forwarded without packet loss.

VLAN

The S5735-S-V2 switch supports 4,094 VLANs.

ACL

The S5735-S-V2 switch supports 2,432 IPv4 ACL rules, all of which can successfully match traffic and execute the configured actions (e.g., blocking).

Port Buffer

The S5735-S-V2 switch supports 0.66MB port buffer.

Routing Protocol

IPv4/IPv6 Routing Protocol

The S5735-S-V2 switch supports IPv4 routing protocols such as RIP, OSPF, IS-IS and BGP, as well as IPv6 routing protocols including RIPng, OSPFv3, IS-IS for IPv6, and BGP4+.

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S5735-S-V2 Series
Switches

Performance
Evaluation and
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Tested
March
2024

Loop Prevention Protocol

STP/RSTP/MSTP

S5735-S-V2 supports STP, RSTP and MSTP to prevent loops.

ERPS

S5735-S-V2 supports Ethernet Ring Protection Switching (ERPS).

Cloud Management

Through a cloud management platform, the S5735-S-V2 switch enables cloud-based configuration, monitoring, and inspection, reducing the need for on-site deployment and operational manpower, thereby lowering network OPEX. Additionally, the switch supports both local management and cloud-managed modes, facilitating smooth evolution and protect user investments.

NETCONF

The S5735-S-V2 switch in cloud-managed mode can be managed in the cloud by the Huawei iMaster NCE-Campus controller through the NETCONF protocol.



Plug-and-play

When used with the Huawei iMaster NCE-Campus controller, the S5735-S-V2 switch supports plug-and-play. New devices are automatically managed by the controller once they are powered on and connected to the network.

Intelligent Upgrade

The S5735-S-V2 switch supports integration with the Huawei Online Upgrade Platform (HOUP), enabling intelligent upgrades via Internet or the iMaster NCE-Campus controller.

Security

CPU Defense - CPCAR

Certain types of protocol packets, including ARP requests, ICMP, DHCP Discover, and others, are forwarded to the switch's CPU for processing. The switch's security features are designed to prevent CPU overload and protect against attacks.

Control Plane CAR (CPCAR) restricts the rate at which protocol packets are sent to the control plane. The switch can limit the rate based on protocols or restrict user-defined flows using ACLs. The S5735-S-V2 switch supports CPU attack prevention through CPCAR.

CPU Defense - Blacklist

The S5735-S-V2 switch supports CPU defense blacklist. Protocol packets matching the source address on the blacklist are blocked.

CPU Defense - Attack Source

Tracing

Tolly engineers verified three attack source tracing features of the S5735-S-V2 switch.

Whitelisting: The switch does not trace whitelisted users, ensuring that legitimate protocol packets from whitelisted users can be sent to the CPU for processing.

Attack source tracing: Administrators can set the threshold and sampling ratio for attack source tracing. When the number of protocol packets sent from an attack source in a specified period exceeds the threshold, the switch traces and logs the attack source to notify the administrator and perform attack source punishment.

Attack source punishment: Administrators can configure attack source punishment to discard or shut down the interface when an attack source is traced. Engineers verified these functions by setting triggering attacks and confirming that the attack was detected at the configured level and that the switch port was disabled once the attack was detected.

DHCP Snooping

The S5735-S-V2 switch supports DHCPv4 snooping and DHCPv6 snooping to create DHCP dynamic binding tables, which are used for security features such as IP Source Guard (IPSG), DAI and SAVI as introduced below. Also, only the DHCP server connected to the DHCP snooping trusted port can distribute IP addresses to the network.

IPSG

IP Source Guard is a security feature that restricts IP traffic on untrusted Layer 2 ports by filtering traffic based on the DHCP user-bind table (static or created by DHCP snooping).

After the IP or MAC address of a host was manually changed to not match the DHCP user-bind table, Tolly engineers verified that the host's IP traffic was all discarded by the S5735-S-V2 switch.

DAI

The S5735-S-V2 switch supports Dynamic ARP Inspection (DAI). ARP packets have to match the DHCP user-bind table (static or created by DHCP snooping) on IP, MAC, VLAN and interface to be forwarded.

SAVI

With the Source Address Validation Improvements (SAVI) feature, the S5735-S-V2 switch is able to check the validity of the source addresses in the Neighbor Discovery (ND) packets, DHCPv6 packets, and IPv6 data packets. The S5735-S-V2 switch is able to filter out invalid packets based on the user-bind table. The user-bind table is generated by ND snooping and DHCPv6 snooping. To check the validity of the source addresses in IPv6 data packets, the IP source guard feature needs to be enabled.

MACsec

MACsec defined by the IEEE 802.1AE standard provides security for Layer 2 networks. All ports on the S5735-S-V2 switch support MACsec to provide encryption and other security measures for Layer 2 traffic.

Trusted Boot

S5735-S-V2's boot file has the digital signature. The device can only boot from legit boot files.

User Authentication

802.1X/MAC Authentication

The S5735-S-V2 switch can work as the authentication policy enforcement point to implement 802.1X and MAC authentication for users.

NAC User Capacity

S5735-S-V2 supports up to 10,000 concurrent authenticated users.



Free Mobility

The S5735-S-V2 switch supports free mobility based on user control list (UCL) groups. With a UCL group, an administrator can create an ACL rule, "rule 1 deny IP source UCL-group name Test destination [File Server's IP address]", to dynamically prevent users in the "Test" UCL group from accessing the file server. Each user is granted the same network access permission regardless of whether the user is a wired or wireless user, where the user logs in, and which IP address the user obtains.

Mirroring

Local Port Mirroring

The S5735-S-V2 switch supports local port mirroring, which involves mirroring the traffic from specified ports to other designated observation ports.

High Availability

BFD

The S5735-S-V2 switch supports Bidirectional Forwarding Detection (BFD) to monitor network connectivity.

Smart Link

The S5735-S-V2 switch supports Smart Link dual homing network, with failover time tested to be less than 15ms.

Multicast

IPv4/IPv6 Layer 2 Multicast

The S5735-S-V2 switch supports Layer 2 multicast for IPv4 and IPv6 based on IGMP snooping and MLD snooping.

Multicast VLAN

The S5735-S-V2 switch supports multicast VLAN. After deploying multicast VLAN on

the Layer 2 device, the upstream router does not need to replicate multicast streams for each user VLAN. Instead, the data stream is replicated within the multicast VLAN and then sent to the Layer 2 device, thus saving network bandwidth and reducing the burden on upstream routers.

PoE

The S5735-S-V2 PoE models supports PoE to provide power to other devices.

Other Features

SNMP

S5735-S-V2 supports SNMP to interoperate with network management systems.

sFlow

The S5735-S-V2 switch supports sFlow to monitor flow statistics.

Telemetry

The S5735-S-V2 switch supports Telemetry, allowing network analyzers to monitor device status and statistics in real-time.

Network Slicing

The S5735-S-V2 switch supports network slicing, providing mutual isolation between slices and ensuring slice bandwidth. Congestion in traffic outside the slice does not affect the guaranteed bandwidth of the slice.

Real-time Clock (RTC)

S5735-S-V2 supports Real-time Clock (RTC).

Last Word of the Power Module

The S5735-S-V2 switch supports the last word feature for power modules. When a power module experiences an abnormality, it leaves behind final fault information for administrators to diagnose.

802.3az EEE

The S5735-S-V2 switch supports 802.3az Energy Efficient Ethernet (EEE). It automatically saves energy when ports are idle and have no traffic flowing through them.

SVF

Super virtual fabric (SVF) is a vertical virtualization technology that allows low-cost fixed switches to act as remote interface cards for a parent switch. This technology increases the port density of the parent switch and facilitates centralized switch management, allowing for high-density access and simple management.

The S5735-S-V2 switch supports working as the AS device (remote interface card) of the SVF.

SZTP

SZTP refers to Secure Zero Touch Provisioning in DHCP mode, where a Bootstrap server is deployed to ensure the trustworthiness of ZTP data through mutual authentication and data encryption. The S5735-S-V2 switch supports SZTP.

Configuration Rollback

The S5735-S-V2 switch supports configuration rollback, allowing the device's configuration to be reverted to a previously saved configuration file.

QoS Rate Limit

The S5735-S-V2 switch supports configuring QoS to limit the traffic rate of specified flows.



Python Script

The S5735-S-V2 switch supports running Python scripts. For example, administrators can use this feature to customize monitoring functions for device status.

HTM

The S5735-S-V2 has the hardware trust module (HTM) installed. Administrators can check the status of the module.

Test Methodology

Each capacity is tested individually during the capacity testing.



Huawei CloudEngine S5735-S-V2 Series Switches

Tolly Verified Features - Part 1 of 2

Ports		Cloud-managed Mode	
✓	GE, Multi-GE (up to 2.5GE), and 10GE ports	✓	Managed by cloud management platforms (e.g. Huawei iMaster NCE-Campus in public or private cloud) via NETCONF
Device Specifications		✓	Plug-and-play with the Huawei iMaster NCE-Campus controller
✓	MAC Table: 32,768 MAC addresses	✓	Intelligent Upgrade with the Huawei Online Upgrade Platform (HOUP)
✓	ARP Table: 4,000 ARP entries	Security	
✓	ND Table: 3,072 ND entries	✓	CPU defend policy - CPCAR Device level rate limit for traffic of certain protocols (e.g. ICMP, ARP, etc.) to protect the CPU
✓	Routing Table: 8,192 IPv4 routes 3,072 IPv6 routes	✓	CPU defend policy - blacklist
✓	FIB: 8,192 IPv4 routes 3,072 IPv6 routes	✓	CPU defend policy - attack source tracing Interface level feature. Identify the attacker and respond with certain actions (interface error down, alarm, etc.)
✓	VLAN: 4,094 VLANs	✓	DHCP snooping and DHCPv6 snooping
✓	ACL: 2,432 IPv4 ACL rules	✓	IP Source Guard (IPSG) Prevent IP address spoofing attacks (unauthorized hosts access and attack the network with forged IP addresses). The device validates IP packets' source IP, source MAC, VLAN ID and interface with the binding table (static or DHCP snooping)
✓	0.66MB port buffer	✓	Dynamic ARP Inspection (DAI) Prevent man-in-the-middle attacks and theft on authorized users' information. The device validates ARP packets' source IP, source MAC, VLAN ID and interface with the binding table (static or DHCP snooping)
Routing Protocols		✓	Source Address Validation Improvements (SAVI) Validate DHCPv6, ND and IPv6 packets with the binding table
✓	IPv4 Routing Protocols: RIP, OSPF, IS-IS, BGP	✓	MACsec
✓	IPv6 Routing Protocols: RIPng, OSPFv3, IS-IS for IPv6, BGP4+	✓	Trusted boot with the digital signature
Loop Prevention Protocols			
✓	STP/RSTP/MSTP		
✓	ERPS		

Source: Tolly, March 2024

Table 1



Huawei CloudEngine S5735-S-V2 Series Switches Tolly Verified Features - Part 2 of 2

User Authentication		Other Features	
✓	802.1X RADIUS authentication	✓	SNMP
✓	MAC authentication	✓	sFlow
✓	Web Portal authentication	✓	Telemetry monitoring
✓	Free Mobility with consistent user access permission regardless of the user location	✓	Network slicing
✓	NAC capacity: 1,024 concurrent authenticated users	✓	Real-time Clock (RTC)
Mirroring		✓	Last word of the power module when it fail
✓	Local Port Mirroring	✓	802.3az Energy Efficient Ethernet (EEE)
High Availability		✓	Super virtual fabric (SVF) as the AS device
✓	BFD	✓	Secure Zero Touch Provisioning (SZTP)
✓	Smart Link Less than 15ms failover time	✓	Configuration rollback
Multicast		✓	QoS rate limit
✓	IPv4/IPv6 Layer 2 multicast	✓	Python scripts
✓	Multicast VLAN	✓	Hardware Trust Module (HTM)
PoE			
✓	Some models support Power over Ethernet (PoE) to power other devices		

Source: Tolly, March 2024

Table 2



About Tolly

The Tolly Group companies have been delivering world-class ICT services for more than 35 years. Tolly is a leading global provider of third-party validation services for vendors of ICT products, components and services.

You can reach the company by E-mail at sales@tolly.com, or by telephone at +1 561.391.5610.

Visit Tolly on the Internet at:
<http://www.tolly.com>

Test Equipment Summary

Vendor	Product	Web
Huawei	CloudEngine S5735-S-V2 Huawei Yunshan OS Version 1.23.0.1 V600R023C00	 HUAWEI https://e.huawei.com
Spirent	TestCenter	 spirent™ Promise. Assured. https://www.spirent.com

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Test Results

Tolly engineers tested functions and performance of Huawei CloudEngine S5735-S series switches (hereinafter referred to as the S5735-S switch). Test results apply to all Huawei CloudEngine S5735-S models as listed in Figure 2. For summary of the test cases, refer to Table 2 and Table 3 on pages 6 and 7. Test results are as follows.

Port Capability

Port Performance

The 10/100/1000BASE-T ports and 10GbE SFP+ optical ports on the S5735-S switch support line-rate forwarding of traffic with different frame sizes, as described in Table 1.

Device Capacity

MAC Table Capacity

The S5735-S switch supports 16K (16,384) MAC addresses in its MAC table. Tolly engineers verified that the switch forwarded

traffic matching all entries in the MAC table, without frame loss or broadcasts occurring.

ARP Table Capacity

The S5735-S switch supports 8,000 entries in its ARP table. Tolly engineers verified that the switch forwarded traffic matching all entries in its ARP table, without any packet loss.

Routing Table/FIB Capacity

The S5735-S switch supports 8K IPv4 routes in both its IPv4 routing table and FIBv4 table. Tolly engineers verified that the switch forwarded traffic matching all routing entries in the FIBv4 table, without any packet loss.

The S5735-S switch supports 3K IPv6 routes in both its IPv6 routing table and FIBv6 table. Tolly engineers verified that the switch forwarded traffic matching all routing entries in the FIBv6 table, without any packet loss.

VLAN Capacity

The S5735-S switch supports 4,094 VLANs.

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CloudEngine
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Performance
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Tested
November
2019

ACL Capacity

The S5735-S switch supports 2,432 ACL rules. Tolly engineers verified that all ACL rules worked properly to match traffic and perform configured actions (e.g. deny).

Huawei CloudEngine S5735-S Series Switch Performance (% of Line-rate)
(as reported by Spirent TestCenter)

Frame Size (Bytes)	64	128	256	512	1024	1280	1518
GbE Ports	100%	100%	100%	100%	100%	100%	100%
10GbE Ports	100%	100%	100%	100%	100%	100%	100%

Notes: 100% line-rate bidirectional traffic between two ports ("GbE to GbE" or "10GbE to 10GbE") was used with zero frame loss. The CloudEngine S5735-S24P4X model was used as the device under test.

Source: Tolly, November 2019

Table 1



Loop-free/Ring Protocols

STP/RSTP/MSTP

The S5735-S switch supports the Spanning Tree Protocol (STP), Rapid Spanning Tree Protocol (RSTP), and Multiple Spanning Tree Protocol (MSTP).

ERPS

The S5735-S switch supports the Ethernet Ring Protection Switching (ERPS) protocol with less than 50ms failover time.

SEP

The S5735-S switch supports the Smart Ethernet Protection (SEP) protocol with less than 50ms failover time.

RRPP

The S5735-S switch supports the Rapid Ring Protection Protocol (RRPP) with less than 50ms failover time and less than 7ms fallback time.

Routing Protocols

The S5735-S switch supports IPv4 routing protocols such as RIP, OSPF, IS-IS and BGP, as well as IPv6 routing protocols such as RIPng, OSPFv3, IS-IS for IPv6 and BGP4+.

Security

Certain types of protocol packets including ARP requests, ICMP, DHCP Discover, etc. are sent to a switch's CPU for processing. It's critical that the switch provides certain attack defense features to prevent CPU overload.

CPU Attack Defense

Two functions of CPU Attack Defense were verified on the S5735-S switch by Tolly engineers.

Blacklist - Administrators can create a blacklist by defining an ACL. Then the switch discards any protocol packets matching the ACL rules.

CPCAR - Control Plane Committed Access Rate (CPCAR) limits the rate of protocol packets sent to the control plane. The switch can limit the traffic rate based on either the protocol type or ACL.

Attack Source Tracing

Three functions of Attack Source Tracing were verified on the S5735-S switch by Tolly engineers.

Whitelist - The switch does not trace the source of users in the whitelist, ensuring that valid protocol packets from users in the whitelist can be sent to the CPU for processing.

Attack source tracing - Administrators can set the threshold and sampling ratio for attack source tracing. When the number of protocol packets sent from an attack source in a specified period exceeds the threshold, the switch traces and logs the attack source to notify the administrator and perform attack source punishment.

Attack source punishment - Administrators can configure attack source punishment to discard or shut down the interface when an attack source is traced.

MFF

MAC-forced Forwarding (MFF) isolates user devices in a broadcast domain at Layer 2. MFF ensures that all traffic, including traffic in the same VLAN, is sent to the gateway, so

that the gateway can monitor data traffic and prevent malicious attacks between users. The S5735-S switch supports MFF.

IPSG

IP Source Guard is a security feature that restricts IP traffic on untrusted Layer 2 ports by filtering traffic based on the DHCP user-bind table (static or created by DHCP snooping).

After the IP or MAC address of a host was manually changed to not match the DHCP user-bind table, Tolly engineers verified that the host's IP traffic was all discarded by the S5735-S switch.

DAI

The S5735-S switch supports Dynamic ARP Inspection (DAI). ARP packets have to match the DHCP user-bind table (static or created by DHCP snooping) on IP, MAC, VLAN and interface to be forwarded.

DHCPv6 Snooping

The S5735-S switch supports the DHCPv6 snooping feature to make sure that only the DHCPv6 server connected to the trusted ports can distribute IPv6 addresses. It also creates the DHCP user-bind table to record the mapping of each client's IPv6 address, MAC addresses, VLAN and port.

ND Snooping

Neighbor Discovery (ND) snooping is a security feature of IPv6 ND and applies to Layer 2 networks. It creates the ND snooping user-bind table to record the mapping of source IPv6 addresses, MAC addresses, VLANs, and inbound ports of Neighbor Solicitation (NS) packets from IPv6 hosts. Tolly engineers verified that the S5735-S switch supported ND snooping.



SAVI

With the Source Address Validation Improvements (SAVI) feature, the S5735-S switch is able to check the validity of the source addresses in the Neighbor Discovery (ND) packets, DHCPv6 packets, and IPv6 data packets. The S5735-S switch is able to filter out invalid packets based on the user-bind table. The user-bind table is generated by ND snooping and DHCPv6 snooping. To check the validity of the source addresses in IPv6 data packets, the IP source guard feature needs to be enabled.

PPPoE+

PPPoE+, also called PPPoE Intermediate Agent is deployed on the switch that is located between the PPPoE client and the PPPoE server. It binds the user authentication information with the interface information to provide security for PPPoE access.

Tolly engineers verified that the S5735-S switch supported PPPoE+.

Secure Boot

Secure boot is the cornerstone of a secure system and secure storage. It ensures that the program to be run at each boot stage is a trusted one that has not been modified. Huawei uses the secure CPU, eFuse, and other security measures to ensure the boot security of the system. Starting from the hardware trust anchor, Huawei validates each step in the boot process. The system cannot boot if any boot step fails the validation process. Tolly engineers verified that a modified or forged digital signature image file cannot boot the system. The S5735-S switch reported a CRC error or signature error based on the modification type.

Authentication

802.1X/MAC/Web Portal Authentication

The S5735-S switch can work as the authentication policy enforcement point to implement 802.1X authentication, MAC authentication, and web portal authentication for users. The S5735-S switch can support up to 1,024 concurrent online users.

Device Management

Zero Touch Provisioning (ZTP)

The S5735-S switch can work with Huawei eSight Unified Management System, to implement zero touch provisioning (ZTP).

Super Virtual Fabric (SVF)

On a traditional campus network, a large number of access devices are widely distributed and have similar configurations. If these devices are configured and managed via traditional methods, a large amount of work is repeated. Huawei's SVF technology virtualizes core/aggregation devices and access devices (including access switches for wired access and WLAN APs for wireless access) into one logical device. The SVF parent (core/aggregation device) manages and configures SVF clients (access devices), simplifying network management and configuration.

An SVF system's clients support two layers of Access Switches (ASes) and one layer of WLAN APs, with ASes supporting stacked devices.

Tolly engineers verified that the S5735-S switch was able to function as the AS node in an SVF system.

Intelligent Upgrade

The S5735-S switch can be connected to Huawei Online Upgrade Platform (HOUP) to implement intelligent upgrade.

PoE

CloudEngine S5735-S series PoE models (e.g. CloudEngine S5735-S24P4X and other P models) support acting as the PoE power sourcing equipment (PS) to provide power to powered devices (PDs), such as WLAN APs.

Perpetual PoE

PoE-capable S5735-S series switches support perpetual PoE. During soft reboot by command or firmware update, the PoE ports of the switch continuously supply power to powered devices (PDs).

Fast PoE

After the S5735-S switch is powered on, the PoE ports on it start to provide PoE power to powered devices (PDs) within 4 seconds.

802.3af and 802.3at

PoE-capable S5735-S series switches support 802.3af (PoE) and 802.3at (PoE+) standards.

Test Methodology

Capacity

In the capacity test, each item was tested independently.

Huawei CloudEngine S5735-S Series Switches



CloudEngine S5735-S24P4X



CloudEngine S5735-S48P4X



CloudEngine S5735-S24T4X



CloudEngine S5735-S48S4X



CloudEngine S5735-S32ST4X



CloudEngine S5735-S48T4X

Source: Tolly, November 2019

Figure 1



Huawei CloudEngine S5735-S Series Switches

Tolly Verified Features - Part 1 of 2

Interface Capability		Routing Protocol	
✓	GbE Ports Port-to-port Line-rate Forwarding (64- to 1518-Byte RFC2544 standard frame sizes)	✓	RIP
✓	10GbE Ports Port-to-port Line-rate Forwarding (64- to 1518-Byte RFC2544 standard frame sizes)	✓	OSPF
Device Capacity		✓	IS-IS
✓	MAC table: 16K MAC addresses	✓	BGP
✓	ARP table: 8,000 entries	✓	RIPng
✓	Routing table: 8K IPv4 routes 3K IPv6 routes	✓	OSPFv3
✓	FIB: 8K IPv4 forwarding entries 3K IPv6 forwarding entries	✓	IS-IS for IPv6
✓	VLAN: 4,094 VLANs	✓	BGP4+
✓	ACL: 2,432 ACL rules		
Loop-free/Ring Protocol			
✓	STP/RSTP/MSTP		
✓	Ethernet Ring Protection Switching (ERPS) with less than 50ms failover convergence time		
✓	Smart Ethernet Protection (SEP) with less than 50ms failover convergence time		
✓	Rapid Ring Protection Protocol (RRPP) with less than 50ms failover convergence time and 7ms failback convergence time		

Source: Tolly, November 2019

Table 2

Huawei CloudEngine S5735-S Series Switches

Tolly Verified Features - Part 2 of 2

Security		Authentication (as the Network Access Control - NAC Policy Enforcement Point)	
✓	CPU defend policy - CPCAR Device level rate limit for traffic of certain protocols (e.g. ICMP, ARP, etc.) to protect the CPU	✓	802.1X authentication
✓	CPU defend policy - blacklist Device level blacklist to block known attackers	✓	MAC authentication
✓	Attack source tracing Interface level feature. Identify the attacker and respond with certain actions (interface error down, alarm, etc.)	✓	Web authentication (portal authentication)
✓	MAC-Forced Forwarding (MFF) Layer 2 isolation. All Layer 2 communications have to go through the gateway	✓	1,024 concurrent authenticated users
✓	Dynamic ARP Inspection (DAI) Prevent man-in-the-middle attacks and theft on authorized users' information. The device validates ARP packets' source IP, source MAC, VLAN ID and interface with the binding table (static or DHCP snooping)	Device Management	
✓	IP Source Guard (IPSG) Prevent IP address spoofing attacks (unauthorized hosts access and attack the network with forged IP addresses). The device validates IP packets' source IP, source MAC, VLAN ID and interface with the binding table (static or DHCP snooping)	✓	Zero Touch Provisioning (ZTP) with Huawei eSight Unified Management System
✓	DHCPv6 snooping Trusted port for the DHCPv6 server; Binding table creation	✓	Super Virtual Fabric (SVF) Access Switch (AS) role
✓	ND snooping Trusted port for ND; Binding table creation	✓	Intelligent upgrade with the Huawei Online Upgrade Platform (HOUP)
✓	Source Address Validation Improvements (SAVI) Validate DHCPv6, ND and IPv6 packets with the binding table	PoE (P Models only)	
✓	PPPoE+ (PPPoE Intermediate Agent) Add the PPPoE client-side interface information to the PPPoE packets for the BRAS to distinguish between end hosts	✓	Perpetual PoE PoE ports provide continuous power to powered devices during soft reboot (reboot via command line) or firmware update
✓	Secure boot CRC check, signature check and other methods to ensure the switch boots from a legit image	✓	Fast PoE After the switch is powered on, the PoE ports on it start to provide PoE power to PD devices within 4 seconds
		✓	802.3af and 802.3at (PoE and PoE+)

Source: Tolly, November 2019

Table 3



About Tolly

The Tolly Group companies have been delivering world-class ICT services for 30 years. Tolly is a leading global provider of third-party validation services for vendors of ICT products, components and services.

You can reach the company by E-mail at sales@tolly.com, or by telephone at +1 561.391.5610.

Visit Tolly on the Internet at:
<http://www.tolly.com>

Test Equipment Summary

Vendor	Product	Web
Huawei	CloudEngine S5735-S24P4X VRP software, Version 5.170 (S5735 V200R019C00SPC300)	 HUAWEI https://e.huawei.com
Spirent	TestCenter	 spirent™ Promise. Assured. https://www.spirent.com

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